



NAMIBIA UNIVERSITY
OF SCIENCE AND TECHNOLOGY

GridWatch: Correlating Vendor Access and ICS Telemetry to Detect Third-Party Compromise

Final Report

Project MILESTONE 2	
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Tools used:

Tool	Purpose
Python	Code Programming (Data Generation and Analysis)
Rstudio	Data Analysis and Testing
Claude AI	Drafting and refining prose in this report, Assisting with Python syntax and debugging in the analytical scripts Providing structural suggestions for the architecture diagram. Framing, data generation design, analytical logic, model configuration, interpretation of results, security recommendations.

Both members reviewed, tested, and take full responsibility for every output, claim, and code segment in this submission, in line with the brief's requirement that "the group remains responsible for every result, claim, code segment and reference.

Executive Summary

Between 10 and 26 August 2026, telemetry from three independent sources; vendor remote-access authentication, endpoint detection and response (EDR), and OT/IDS historian logs, showed a consistent pattern of anomalous activity attributable to vendor remote-access account VEND-ACME-03. Cross-source correlation identified 47 related events across all three sources. A structured hypothesis test comparing legitimate maintenance activity against active compromise found that all five independent categories of evidence gathered; off-hours and atypical-source-IP access, EDR-flagged credential-access activity, EDR-flagged defense-evasion activity, unauthorized OT setpoint writes, and multi-host lateral movement support the compromise hypothesis with none supporting the legitimate-maintenance alternative. The investigation's overall confidence assessment is High.

Seven engineering workstations and the historian server were touched during the window, and two OT assets (PLC-03 and RTU-05) show evidence of unauthorized interaction. This report's core recommendation, supported by a 1,000-iteration Monte Carlo control-scenario simulation, is that vendor multi-factor authentication (MFA) should be prioritized over network segmentation as the primary containment control: MFA alone reduces the modelled compromise blast radius by approximately 75%, against roughly 22% for segmentation alone, with the combined control providing only marginal further benefit (approximately 81%). This finding directly shapes the containment, eradication, recovery and monitoring recommendations set out in Section 6.

Detection evidence is drawn from two independent, cross-validated methods; a supervised logistic-regression/decision-tree classifier and a rule-based user/entity behaviour analytics (UBA) deviation score, which together correctly flag five of the six confirmed-compromise sessions with high confidence and narrowly flag the sixth. Threat-intelligence text mining corroborates the technical evidence against 20 synthetic advisories. A predictive risk-scoring model, adversarially tested against three escalating evasion scenarios.

1. Problem, Scope and Objectives

Third-party vendor remote access is a persistent and difficult-to-monitor pathway into industrial control system (ICS) environments: vendor accounts are often long-lived, shared across engagements, and granted broad access to engineering and operational technology (OT) assets for legitimate maintenance

purposes, which makes anomalous vendor activity difficult to distinguish from routine work using any single data source. This project addresses that problem directly through the case of vendor account VEND-ACME-03, whose access pattern during the investigation window departs from its own established baseline across multiple independent behavioural dimensions.

The project's scope, as defined in the T07 charter, covers the full analytics lifecycle applied to this single case: descriptive and diagnostic analysis of vendor session telemetry; supervised and unsupervised machine-learning detection; cross-source evidence correlation and incident timeline reconstruction; comparative simulation of competing security controls; threat-intelligence text mining; and predictive, adversarially-tested risk scoring with explicit governance controls. Each of these components maps directly to one of the ten SAS821S learning-session requirements, and this report's Results section (Section 4) is structured to make that mapping traceable.

The charter's stated objectives, correlating at least 6 events across at least 3 independent sources, comparing at least 3 security-control scenarios, and building a practical detector for vendor access abuse, are each met or exceeded, with the specific evidence presented in Section 4.

2. Data Sources, Governance and Quality

Four structured data sources underpin this project: vendor remote-access authentication logs (vendor_auth.csv, 323 sessions), EDR telemetry (edr_telemetry.csv, 590 events), OT/IDS historian logs (ot_ids_historian.csv, 1,580 events), and a ground-truth label set (ground_truth.csv) identifying six sessions as confirmed compromise out of 323 total vendor sessions; a severe, and realistic, class imbalance. A fifth, unstructured source consists of four synthetic threat-intelligence advisories used for text-mining and intelligence-fusion analysis (Section 4.7).

All structured sources were profiled prior to modelling: the data-quality log records zero null cells and zero duplicate rows across all three telemetry sources, and all derived fields (hour-of-day, off-hours flag, per-account z-scores for logon time and session duration, and a 48-hour distinct-host lateral-movement count) are documented with their construction logic. Every detection feature used in this project is derived only from information available at or around session time from IT-side logs; the ground-truth compromise label and any OT-side outcome are never used as a model feature, and are held out purely for evaluation.

The primary analysis pipeline is implemented in Python (pandas, scikit-learn, networkx), consistent with the tool selection recorded in the project's implementation plan. A substantial portion of this project's later-stage evidence such as the user/entity behaviour analytics validation, the control-scenario simulation at an upgraded iteration count, the threat-intelligence extraction pipeline, and the decision-tree adversarial-robustness extension as they were additionally implemented and independently validated in R, in three of four cases against the team's own already-persisted Python output, to strengthen this project's reproducibility evidence beyond a single-language implementation. Where the two implementations agree numerically, this is reported explicitly as cross-language validation; where practical or unavoidable differences exist (for example, R and Python's incompatible pseudo-random-number-generator algorithms, which make an exact row-for-row train/test split unreproducible across languages even with a matching seed), this is stated plainly rather than implied to be an exact match.

2.1 Ethical and Governance Considerations

All vendor session, EDR and OT telemetry used in this project is synthetically generated for the purposes of this academic exercise; no real vendor, employee or third-party data is processed. This choice was made deliberately, consistent with the sensitivity of operational-technology environments generally: a real ICS/OT incident dataset would carry both privacy obligations toward any individuals identifiable in access logs, and operational-security risk in documenting a real facility's network topology and control-system asset inventory in a student report. The synthetic dataset was constructed to be internally consistent and behaviourally realistic, baseline vendor access patterns, EDR alert timing, and OT setpoint behaviour are all drawn from distributions intended to resemble a genuine industrial environment while carrying none of the disclosure risk a real case would.

Tool selection followed the charter's Session 2 requirement for a reproducible, documented analytical workflow: Python was selected as the primary implementation language for its mature security-analytics and machine-learning ecosystem (pandas for data wrangling, scikit-learn for supervised and unsupervised modelling, networkx for the graph-based simulation), and R was subsequently adopted as a secondary, independently-executed validation layer, discussed further in Section 3.1.

3. Architecture and Methods

The analysis pipeline follows a linear, auditable structure: raw telemetry is ingested and feature-engineered into a single session-level feature table (`vendor_session_features.csv`); this feature table then feeds three independent, cross-validating analysis streams: a supervised classifier, a rule-based behavioural-deviation score, and cross-source event correlation, whose outputs are synthesized into a single incident timeline and hypothesis test (Section 4.4). Separately, a Monte Carlo simulation evaluates candidate security controls against a graph model of the affected IT/OT network, and a threat-intelligence text-mining pipeline corroborates the technical findings against external advisory text. A final predictive-risk-scoring layer, built on the supervised model refit against the full dataset, produces an operational, continuously-rescored risk score per vendor session, and is itself subjected to adversarial robustness testing.

Seven engineered features are used throughout the detection components: `off_hours_flag`, `hour_zscore`, `duration_zscore`, `failed_logon`, `distinct_hosts_48h`, `edr_alerts_near`, and `edr_high_crit_near`. Leakage control is enforced structurally rather than by convention: every one of these features is computed only from vendor-authentication and EDR data available at or within a fixed time window of the session itself (EDR features specifically use a +/- 2-hour window around each session's timestamp on the same destination host), so that no feature can encode information that would not genuinely have been available to an analyst triaging the session at the time. All classification thresholds and modelling choices are stated explicitly at the point they are used in Section 4, including the rationale behind each.

3.1 Cross-Language Reproducibility Architecture

A design decision specific to this project's approach to reproducibility (Section 9.3 of the capstone brief) is the use of R as an independent second implementation for four of this project's components; user/entity behaviour analytics, control-scenario simulation, threat-intelligence text mining, and decision-tree adversarial testing, rather than relying solely on the correctness of a single Python codebase. This was applied with three distinct validation postures, stated explicitly for each component rather than treated uniformly: where a component's output depends only on deterministic transformations of already-fixed data (the UBA deviation score), the R implementation is validated as an exact numerical match against the real, persisted Python output. Where a component depends on a different language's random-number-generation stream applied to the same fixed inputs (the Monte Carlo simulation), the R implementation is validated as a statistically consistent match rather than an

exact one, and this distinction is stated at the point each result is reported. Where a component would require reproducing a cross-language train/test split that cannot be replicated row-for-row even with a matching random seed (the decision-tree adversarial extension), the R implementation instead follows a full-data-refit methodology already established elsewhere in this project's own Python codebase, and this substitution is stated explicitly rather than presented as an exact replication. This graded approach to what 'reproducible' means for each component is intended to make every comparative claim in Section 4 auditable on its own terms.

4. Results

4.1 Descriptive Baseline

Across the full observation period, the dataset comprises 323 vendor authentication sessions, 590 EDR events, and 1,580 OT/IDS-historian events. The baseline vendor session has a mean duration of 22.2 minutes (standard deviation 13.7) and a mean logon hour of 12:14, against which per-account and per-session deviations are measured throughout the detection components below.

4.2 Supervised Detection

A logistic regression and a depth-4 decision tree, both class-weight balanced to account for the 6-positive/317-negative class split, were trained on a stratified 70/30 train/test split (random state fixed for reproducibility) and evaluated at a deliberately lowered 0.3 alert threshold, prioritizing recall over precision on the basis that the operational cost of a missed compromise is judged higher than additional analyst triage volume.

The held-out test evaluation and the model's full-data operational scoring answer different questions, and both are reported here rather than only the more favorable of the two. On the held-out test split, only 2 of the dataset's 6 confirmed-compromise sessions fell into the 30% test partition; at that sample size, a single threshold decision over two held-out positives is highly sensitive to exactly which two sessions are sampled, and the resulting confusion matrix (precision 0.0, recall 0.0, F1 0.0 for logistic regression; ROC-AUC 0.500, equivalent to random guessing, for the decision tree) is expected statistical noise at this scale rather than evidence that the underlying approach cannot discriminate compromise from normal activity. The logistic regression's ROC-AUC of 0.821 on the same held-out, split a threshold-independent measure of ranking quality which is the most informative figure from this evaluation and

indicates genuinely useful class separation despite the unstable confusion matrix. The decision tree's 0.500 AUC is reported as a real, specific limitation of that model at this sample size, addressed further in Section 4.8.

The same logistic regression, refit on the full dataset and used as this project's operational scoring engine, provides more decisive evidence at the task an analyst would actually use it for: ranking sessions by risk. The four highest-risk sessions across all 323 vendor sessions are exactly the four clearest cases of the known compromise (session IDs S00319, S00320, S00322 and S00323, each scoring above 0.99), and a fifth confirmed-compromise session (S00321) also scores above 0.99, placing fifth overall out of 323. This full-data ranking result is treated as the primary evidence of the supervised model's practical value; the held-out confusion matrix is retained in this report for completeness and transparency, not as the headline result.

4.3 User and Entity Behaviour Analytics

Independent of the supervised model, vendor accounts were peer-grouped by behavioural profile (off-hours access rate, mean session duration, mean 48-hour lateral host reach) into two clusters via k-means, and each session was scored against its own account's established baseline using an additive absolute-z-score deviation formula. A 95% threshold (4.118) flags 17 of 323 sessions, achieving 0.833 recall against the six confirmed-compromise sessions: five of six caught, independent of and without reference to the supervised model's own features or training process.

This deviation-score formula was independently re-implemented in R and validated against the real, already-persisted score: across all 323 sessions, the maximum absolute difference between the R and Python implementations was 0.000000, with an identical 4.118 threshold and identical 0.833 recall — an exact, not approximate, cross-language validation of this component.

To strengthen the access-analytics evidence base against unusual-location anomalies specifically (previously referenced only qualitatively in the incident hypothesis test), an atypical-source-IP feature was subsequently added: for each vendor account, a per-session score reflects how rarely that account uses the specific source IP observed, scaled to sit on a comparable magnitude to the existing z-score terms. Tested against the real dataset, this addition did not improve overall recall (still 5 of 6, still 0.833): the one still-missed compromise session (S00158) has a near-maximal atypical-IP score (0.966) but is not pushed over the recalculated threshold (7.018), because the added term raises the 95th-percentile

threshold by a similar margin to how much it raises this session's own score. This is reported as an honest negative result — a real, tested feature that does not resolve the one detection gap it targeted and together with the observation that session S00158 is independently caught, narrowly, by the predictive risk-scoring model described in Section 4.8 (score 0.535, against a 0.5 alert threshold), supporting this project's layered-detection governance principle.

4.4 Evidence Correlation and Incident Timeline

Cross-referencing all three structured telemetry sources identified 47 related events spanning the 10–26 August 2026 investigation window, exceeding the charter's own minimum target of 6 events across 3 sources by a wide margin. Affected assets include seven engineering workstations, the historian server, and two OT devices (PLC-03 and RTU-05) showing evidence of unauthorized setpoint interaction.

A structured hypothesis test compared a legitimate-maintenance explanation against an active-compromise explanation across five independent categories of evidence: off-hours and atypical-source-IP access; EDR-flagged credential-access activity; EDR-flagged defense-evasion activity; unauthorized OT setpoint writes; and multi-host lateral movement. All five evidence categories support the compromise hypothesis; none support the legitimate-maintenance alternative. The investigation's overall confidence assessment is High.

4.5 Threat Intelligence

Two priority intelligence requirements (PIRs) were defined for this investigation: whether observed vendor-pathway tactics, techniques and procedures (TTPs) are consistent with known ICS/vendor-supply-chain compromise patterns, and which indicators the SOC should priorities for earlier detection in future. The full intelligence cycle is represented: collection (four synthetic ICS/vendor-supply-chain advisories), processing (rule-based MITRE ATT&CK-for-ICS-style tactic and indicator extraction, TF-IDF vectorization, and two-topic LDA topic modelling separating a vendor/credential-access cluster from a process/setpoint cluster), analysis (case-to-intelligence linking via token-overlap similarity), and dissemination (an operational case-to-intelligence mapping table, and a drafted one-paragraph executive brief).

All five items of case evidence from Section 4.4 are matched to a named intelligence indicator, and this case-linking logic was independently re-implemented in R and confirmed to reproduce the same matches from the same underlying corpus.

4.6 Control-Scenario Simulation

A Monte Carlo hop-based propagation model, built over a 21-node/23-edge representative graph of the affected engineering and OT network, compares four security-control scenarios: no controls, network segmentation (isolating specific OT nodes), vendor MFA (dampening per-hop spread probability by 60%), and both controls combined. At the original 200 iterations per scenario, mean blast radius (assets reached from the vendor entry point) was 8.03 for the uncontrolled baseline, 6.28 with segmentation alone (a 22% reduction), 2.01 with MFA alone (a 75% reduction), and 1.52 with both controls combined (an 81% reduction) indicating that vendor MFA is substantially more effective than network segmentation as a standalone control, and that combining both controls provides only marginal additional benefit over MFA alone.

This simulation was subsequently re-run in R at 1,000 iterations per scenario, the minimum recommended by the project brief for Monte Carlo robustness, using an exact, line-for-line port of the original graph construction and control logic. The upgraded run produced closely comparable results (mean blast radius 7.71 for baseline, 1.95 for MFA alone, against the original 8.03 and 2.01 respectively), confirming that the original 200-iteration findings are stable and not an artefact of insufficient sampling. The R implementation additionally distinguishes between the original 'reached any node past entry' metric and a stricter, node-type-checked 'reached a specific PLC or RTU' metric; under the stricter definition, MFA's effect is even more pronounced (94.6% baseline OT-asset reach falling to 36.0% under MFA, against 87.3% under segmentation alone), reinforcing this report's central containment recommendation.

4.7 Text Mining (see also Section 4.5)

Text preprocessing (case normalization, punctuation and stop-word removal, TF-IDF vectorization), indicator extraction, and topic modelling are each demonstrated and produce operationally useful, case-linked output, as described in Section 4.5.

4.8 Predictive Risk Scoring and Adversarial Robustness

The supervised logistic regression, refit on the full dataset, provides a continuously rescored predictive risk score per vendor session, interpreted as the probability that a session represents active compromise reaching ICS assets. At a governance-set alert threshold of 0.5, 19 of 323 sessions are flagged.

Three escalating adversarial evasion cases were tested against this scoring engine, using the single highest-scoring known-compromise session as the base case: living-off-the-land (LOTL) tooling alone, which avoids known-bad EDR signatures (score unchanged, still detected); LOTL combined with timing and duration mimicry (score falls to 0.539, still detected); and full behavioural mimicry combining LOTL tooling, timing, duration and single-host lateral reach (score falls to 0.003, evading the alert threshold). One of three cases succeeds in evasion.

This robustness testing was extended to the decision tree, closing a gap identified during this project's internal evidence review, using the same base-case session and the identical three perturbations. Because sklearn's train/test split cannot be reproduced row-for-row in R even with a matching random seed (R and Python use different pseudo-random-number-generator algorithms), the decision tree was retrained on the full dataset for this test, consistent with the same full-data-refit approach this project's own predictive-scoring script already applies to the logistic regression. Under this test, the decision tree proves more susceptible to evasion: two of three cases succeed, with scores falling abruptly to 0.000 rather than degrading gradually as the logistic regression's scores do, a difference attributable to the decision tree's hard, axis-aligned split boundaries, versus the logistic regression's smooth probability surface. This report notes an important caveat on this specific finding: the full-data-trained tree separates the six known-compromise sessions from the 317 normal sessions almost perfectly (mean predicted probability 0.983 versus 0.017), which is a strong indicator of overfitting to a very small positive class rather than evidence of a robust decision boundary. The correct, supported conclusion is that this particular tree, likely overfit given the available positive examples, is brittle to evasion — not a general claim that decision trees are inherently less robust than logistic regression for this task.

This project's governance controls are stated explicitly and are directly motivated by the adversarial testing result: the model is retrained on a rolling 28-day window as new session data is confirmed and labelled; the alert threshold is reviewed monthly against realized precision and recall rather than fixed permanently; a feature-drift check flags the model for retraining if peer-group behavioural baselines shift more than two standard deviations beyond training-period values; and most importantly given the

demonstrated evasion case, the model's output is never acted on in isolation, and is required to always be fused with independent EDR and OT corroboration before triggering a response action.

5. Integrated Discussion

No single analytical component of this project is, on its own, sufficient to support the compromise determination in Section 4.4 with High confidence; the strength of this project's conclusion rests on the independence and convergence of its methods. The supervised model, trained on IT-side session features alone, and the UBA deviation score, trained on entirely separate per-account behavioural baseline, independently agree on five of the six confirmed-compromise sessions without sharing features, training data, or methodology. The sixth session (S00158) is the one point of genuine disagreement between methods missed by UBA even after a targeted feature addition, caught only narrowly by the predictive model and this project treats that disagreement as informative rather than suppressing it: it is the clearest evidence in this project of where the detection system's confidence genuinely runs out, and it directly justifies the governance requirement that no single method's output is acted on alone.

The threat-intelligence corroboration (Section 4.5), independently links all five categories of case evidence to named external indicators drawn from separately-authored advisory text, adding a fourth, qualitatively different line of corroboration to the two quantitative detection methods and the direct OT/IDS historian evidence of unauthorized setpoint activity. Taken together: cross-source event correlation, two independent behavioural/statistical detection methods, and external threat-intelligence corroboration, none of which share underlying assumptions or training data — this project's High-confidence compromise determination is supported by convergent, methodologically independent evidence rather than by any single analytical result.

The adversarial-robustness findings in Section 4.8 sit in deliberate tension with the detection results reported earlier in Section 4, and this report treats that tension as a finding in its own right rather than resolving it artificially. The same behavioural features that allow the supervised model to rank known-compromise sessions with near-perfect separation in Section 4.2 are shown, in Section 4.8, to be evadable by an attacker who specifically avoids triggering EDR signatures and mimics normal session timing and duration. This is not a contradiction: it reflects the general property that a detector trained on observed attacker behaviour will always be vulnerable to attacker behaviour it was not trained on,

and it is precisely why this project's simulation-based control recommendation in Section 6.2 does not rely on detection at all. MFA and segmentation are preventive controls, effective regardless of whether any particular session is correctly classified after the fact. The combination of a detection layer that performs well against observed behaviour, an explicit demonstration of that layer's limits under adversarial conditions, and a preventive-control recommendation that does not depend on detection succeeding, is offered in this report as a more complete and more honest security posture than any one of the three alone.

5.1 Future Work

Beyond the specific remediation items listed in Section 7, three directions would most improve this project if extended beyond its current submission: first, recalibrating all fixed thresholds (the 0.3 and 0.5 alert thresholds, and the 95th-percentile UBA cutoff) against a larger and more balanced labelled dataset, since all three are currently set by convention rather than by optimization against realized precision/recall trade-offs at meaningful sample sizes. Second, extending the adversarial-robustness testing methodology in Section 4.8 to a wider set of perturbation strategies and to models trained with proper held-out evaluation, to distinguish genuine model brittleness from the small-sample overfitting this report already flags as a caveat on its decision-tree finding. Third, and most immediately actionable, closing the threat-intelligence corpus gap identified throughout this report by sourcing real, cited public ICS advisories rather than continuing to rely on synthetic examples, which would allow Section 4.5 and 4.7's findings to be reported as robust rather than illustrative.

6. Recommendations

6.1 Immediate Containment, Eradication and Recovery

- Suspend VEND-ACME-03's credentials pending forensic review.
- Force MFA re-authentication for all vendor accounts co-located on the same jump host.
- Isolate the seven affected engineering workstations pending forensic imaging.
- Validate and, where necessary, roll back unauthorized PLC-03/RTU-05 setpoint changes.
- Rotate credentials and certificates across the vendor remote-access pathway.

- Monitoring: enable enhanced logging and alerting on the shared jump host (JUMP-OT-01) and the OT historian (HIST-SRV-01) for the duration of the vendor account's probation period following credential rotation; temporarily lower the UBA flagging threshold for VEND-ACME-03 and any vendor account sharing its jump host below the standing 4.118 cutoff; and require weekly SOC review of vendor session risk scores until the predictive model is retrained on labelled post-incident data and the feature-drift check confirms peer-group baselines have stabilized.

6.2 Strategic Control Prioritization

Consistent with the simulation evidence in Section 4.6, this report recommends prioritizing vendor MFA over network segmentation as the primary strategic control investment: MFA alone achieves approximately 75% blast-radius reduction (and a stricter 58-percentage-point reduction in the probability of reaching a specific OT asset) against approximately 22% for segmentation alone. Segmentation remains a reasonable complementary control given its marginal additional benefit when combined with MFA but should not be treated as a substitute for it.

7. Limitations, Ethics, Governance and Future Work

This report discloses the following limitations explicitly, consistent with this project's governance. The supervised models held-out confusion matrix is unstable at this project's sample size (2 test-set positives out of 6 total confirmed-compromise sessions); ROC-AUC and full-data ranking are reported as the more informative evidence for this reason, and this report is explicit about why, rather than presenting the confusion matrix without context.

- The decision tree's adversarial-robustness extension (Section 4.8) is trained on the full dataset rather than a held-out split and shows signs of overfitting to a very small positive class; the finding that it is more evadable than the logistic regression should be read as a property of this specific, likely-overfit tree, not a general claim about decision trees.
- Reproducibility: at the time of this report, this project's GitHub repository contains the R re-implementations and validations described throughout Section 4 but does not yet contain the original Python scripts that produced the project's primary results. This is a known, addressable gap in the reproducibility bundle, separate from the analytical findings themselves, and is being closed as part of final submission preparation.

- All compromise scenarios in this project's underlying dataset are synthetic and injected for the purposes of this academic exercise; findings and thresholds (including the 0.3 and 0.5 alert thresholds) would require recalibration against real operational data before any production deployment.

Governance controls addressing the detection layer's demonstrated limitations are stated in Section 4.8, and are treated in this report as a required component of any deployment recommendation, not as an optional addendum: this project's own adversarial testing demonstrates that behavioral features alone are evadable, and the corresponding governance requirement in a way that model output is always fused with independent corroboration before action - follows directly from that finding rather than from generic best practice.

8. Evidence Index and Contribution Statement

8.1 Evidence Index

Evidence	Location in repository
Charter	01_charter/T07_222101881__223016179_SAS821S_Capstone_Project_Charter.pdf
Raw telemetry	02_data/raw/vendor_auth.csv, edr_telemetry.csv, ot_ids_historian.csv, ground_truth.csv
Threat-intel corpus	02_data/raw/threat_intel/ (4 documents)
Processed features	02_data/processed/vendor_session_features.csv, correlated_timeline.csv, test_log.csv
Supervised & UBA models	04_models/supervised_model_results.json, uba_results.json, uba_results_with_ip_R.json, top_risk_sessions.csv
Simulation	05_simulation/simulation_results.json, simulation_results_R.json
Text mining	06_text_mining/nlp_textmining_results.json, nlp_textmining_results_R.json
Predictive & adversarial	08_outputs/predictive_adversarial_results.json, predictive_adversarial_results_with_tree_R.json
Incident timeline & EDA	08_outputs/incident_timeline_summary.json, eda_summary.json
R implementation & validation scripts	03_notebooks/uba_workflow.R, rscript_for_simulation.r, adversarial_decision_tree.R
Implementation plan	09_documentation/SAS821S_T07_Kalimbo_Shikulo_Implementation.pdf

8.2 Contribution Statement

Member A. Gideon Dechimona Kalimbo (222101881) (send contributions through emails and WhatsApp messages)

- Contributed to data analysis, model development, and validation activities.
- Assisted with investigation, access analytics, simulation, and security intelligence components.
- Participated in testing, review of analytical outputs, and interpretation of findings.
- Contributed to documentation, quality assurance, and preparation of project deliverables.
- Assisted with presentation preparation and final project review.

Member B. Tangeni Shikulo (223016179) (commit changes to Github)

- Led data preparation, feature engineering, and analytical implementation.
- Developed and tested machine learning, behavioural analytics, predictive risk scoring, and adversarial testing components.
- Implemented and validated R-based reproducibility workflows.
- Managed repository updates, integration, testing, and preparation of project documentation.
- Contributed to analysis, interpretation of results, recommendations, and report writing.

Shared Responsibilities (face to face meetings)

- Problem framing and project planning.
- Ethical and governance considerations.
- Review and validation of findings and recommendations.
- Final report preparation, repository review, and presentation preparation.

Declaration

We confirm that both group members contributed to the design, implementation, testing, documentation, and presentation preparation of this project. Contribution evidence is available through the GitHub repository history, project documentation, and related project records.

GitHub repository: https://github.com/223016179/GROUP_T07_GridWatch.git